Chapter 15: MCQ Exam Preparation

Key Concepts, Common Misconceptions, and Exam Strategies

Gordon Wright

1 Overview

Before You Begin

This reading is designed to help you consolidate your learning and prepare for the MCQ examination. It is not a substitute for understanding the material — it is a tool for organising what you already know.

This reading covers:

- A review of key concepts from the module
- Common misconceptions and how to avoid them
- Strategies for approaching MCQ questions
- What to expect on exam day

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2 Part 1: Core Concepts Review

2.1 The Scientific Method

The scientific method provides a systematic approach to understanding behaviour:

Key Steps:

1. **Identify a question** — What do you want to understand?
2. **Review existing literature** — What is already known?
3. **Formulate a hypothesis** — Make a specific, testable prediction
4. **Design a study** — Plan your method carefully
5. **Collect data** — Gather observations systematically
6. **Analyse results** — Use statistics to evaluate findings
7. **Draw conclusions** — Interpret what the results mean
8. **Report findings** — Share methods and results transparently

ℹ Exam Tip

Questions may present a research scenario and ask you to identify which step of the scientific method is being described. Focus on the key action: predicting = hypothesis, measuring = data collection, interpreting = conclusions.

2.2 Variables

Variable: Anything that can take different values.

Type	Definition	Example
Independent Variable (IV)	What the researcher manipulates	Word type (abstract vs concrete)
Dependent Variable (DV)	What the researcher measures	Number of words recalled
Confounding Variable	An uncontrolled variable that could explain results	Time of day, participant fatigue
Extraneous Variable	Any variable other than the IV that could affect the DV	Noise levels, room temperature

! Key Distinction

- **IV** = manipulated (what the researcher changes)
- **DV** = measured (what the researcher records)

The DV *depends on* the IV. Ask yourself: "Does [DV] depend on [IV]?"

2.3 Research Designs

Design	Definition	Key Feature
Between-subjects	Different participants in each condition	Compare between groups
Within-subjects	Same participants in all conditions	Compare within individuals
Matched pairs	Participants paired on key variables, then assigned to conditions	Combines features of both
Correlational	Measures relationship between variables without manipulation	No IV, no causation

⚠ Common Confusion

In a within-subjects design, the same person provides data for all conditions (e.g., before AND after).

In a between-subjects design, different people are in each condition (e.g., Group A OR Group B).

2.4 Hypotheses

Type	Definition	Example
Experimental (H1)	Predicts an effect or difference	"Concrete words will be recalled better than abstract words"
Null (H0)	Predicts no effect or no difference	"There will be no difference in recall between word types"
One-tailed	Predicts direction	"...will be *better*"
Two-tailed	Predicts difference without direction	"...will *differ*"

2.5 Sampling

Population: The entire group you want to generalise to.

Sample: The subset you actually study.

Sampling frame: The list from which you draw your sample.

Method	Definition	Strength	Weakness
Random	Every member has equal chance of selection	Unbiased	Often impractical
Stratified	Divide population into subgroups, sample from each	Ensures representation	Requires population information
Convenience	Whoever is available	Easy	May be unrepresentative
Snowball	Participants recruit other participants	Reaches hidden populations	Not random

2.6 Experimental Control

Why control matters: If variables other than the IV differ between conditions, you cannot conclude the IV caused the difference.

Random assignment: Randomly allocating participants to conditions — distributes individual differences evenly across groups.

Counterbalancing: Varying the order of conditions across participants to control for order effects.

Standardisation: Keeping procedures identical across conditions (same instructions, same environment).

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3 Part 2: Statistics Concepts

3.1 Descriptive Statistics

Statistic	What It Tells You	Formula
Mean	Average value	Sum of values / number of values
Standard Deviation	Spread from the mean	How much scores typically differ from the mean
Variance	Average squared deviation	SD squared

> ℹ Exam Tip
>
> You may be asked to calculate a mean or identify what SD represents. Remember: SD tells you about variability — low SD means scores cluster together; high SD means scores are spread out.

3.2 The Normal Distribution

Key properties:

- Symmetrical and bell-shaped
- Mean, median, and mode are equal
- 68% of values within 1 SD of the mean
- 95% within 2 SDs
- 99.7% within 3 SDs

3.3 Inferential Statistics

The goal: Determine whether observed differences are likely due to the IV or could have occurred by chance.

Concept	Definition
p-value	Probability of obtaining results at least this extreme if the null hypothesis were true
Alpha level	The threshold for significance (typically .05)
Significance	If p < alpha, we reject the null hypothesis
Type I error	False positive — rejecting H0 when it is true
Type II error	False negative — failing to reject H0 when it is false

! Critical Understanding

p < .05 does NOT mean:

- There is a 95% chance the hypothesis is true
- The effect is large or important
- The study will replicate

It DOES mean: If H0 were true, results this extreme would occur less than 5% of the time by chance.

3.4 Which Test When?

Comparison	Test	Example
Two groups (between)	Independent t-test	Lab A vs Lab B recall
Two conditions (within)	Paired t-test	Pre vs post mood
Relationship between variables	Correlation	Self-rating and actual accuracy

3.5 Effect Size

Effect size tells you the magnitude of an effect, independent of sample size.

- **Cohen's d**: For differences between means (small = 0.2, medium = 0.5, large = 0.8)
- **Pearson's r**: For correlations (small = 0.1, medium = 0.3, large = 0.5)

A significant result with a tiny effect size may not be practically meaningful.

4 Part 3: Research Credibility

4.1 Reliability and Validity

Concept	Definition	Question It Answers
Reliability	Consistency of measurement	Do you get the same result each time?
Validity	Accuracy of measurement	Are you measuring what you intend to?

Types of reliability:

- **Test-retest**: Same results over time
- **Inter-rater**: Same results across different observers
- **Internal consistency**: Items within a scale measure the same construct

Types of validity:

- **Face validity**: Appears to measure what it should (weakest form)
- **Content validity**: Covers all aspects of the construct
- **Criterion validity**: Predicts or correlates with relevant outcomes
- **Construct validity**: Actually measures the theoretical construct

4.2 Internal vs External Validity

Internal validity: Can you conclude that the IV caused changes in the DV?

- Threatened by confounds, lack of control, order effects

External validity: Can you generalise findings beyond this specific study?

- Threatened by unrepresentative samples, artificial settings

4.3 Replication and Open Science

Key concepts from Week 17:

- **Replication crisis**: Many published findings fail to replicate
- **P-hacking**: Exploiting flexibility to achieve significance
- **HARKing**: Presenting post-hoc hypotheses as predictions
- **Publication bias**: Significant results more likely to be published
- **Pre-registration**: Documenting hypotheses before data collection

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5 Part 4: Common Misconceptions

5.1 Misconception 1: Correlation Equals Causation

⚠ Wrong

"The study found a correlation between screen time and depression, so screen time causes depression."

6 Correct

A correlation shows two variables are related, but does not establish that one causes the other. There could be reverse causation (depression leads to more screen time) or a third variable explaining both.

6.1 Misconception 2: Non-significance Means "No Effect"

⚠ Wrong

"The study found $p = .08$, so there is no relationship between the variables."

7 Correct

Non-significance means we cannot confidently reject the null hypothesis. It does not prove H_0 is true. The study may have lacked power to detect a real effect.

7.1 Misconception 3: p-value Is the Probability the Hypothesis Is True

⚠ Wrong

"$p = .03$ means there is a 97% chance the hypothesis is correct."

8 Correct

The p-value is the probability of the data (or more extreme) given that H_0 is true. It is not the probability that H_1 is true.

8.1 Misconception 4: Random Assignment and Random Sampling Are the Same

⚠ Wrong

"Participants were randomly assigned, so the sample is representative of the population."

9 Correct

Random assignment (allocating to conditions) ensures groups are equivalent.

Random sampling (selecting from population) ensures representativeness.

They serve different purposes and are often confused.

9.1 Misconception 5: Larger Sample Size Makes a Study "Better"

⚠ Wrong

"This study had 10,000 participants, so the findings must be important."

10 Correct

Large samples can detect very small effects. A statistically significant but tiny effect may have no practical importance. Always consider effect size alongside significance.

10.1 Misconception 6: Within-subjects Designs Are Always Better

⚠ Wrong

"Within-subjects designs control for individual differences, so they are always preferable."

11 Correct

Within-subjects designs have their own problems: order effects, practice effects, fatigue, and demand characteristics. The best design depends on the research question.

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12 Part 5: Exam Strategies

12.1 Before the Exam

Effective preparation:

1. **Review actively** — Do not just re-read. Test yourself, explain concepts aloud, create your own examples
2. **Focus on understanding** — If you understand *why*, you can work out the answer even to unfamiliar questions
3. **Know the terminology** — Many questions hinge on precise definitions
4. **Practice with examples** — Apply concepts to the DataLab tasks you completed

12.2 Reading MCQ Questions

Step 1: Read the question stem carefully

- What exactly is being asked?
- Are there key words like "BEST," "ALWAYS," "NEVER," "EXCEPT"?

Step 2: Try to answer before looking at options

- If you know the material, your answer should match one option
- This prevents being misled by plausible-sounding distractors

Step 3: Evaluate all options

- Even if one looks correct, check the others
- Sometimes two options are similar and one is more precise

Step 4: Eliminate clearly wrong answers

- If unsure, narrowing down improves your odds

12.3 Common Question Types

Definition questions: > "Which of the following best defines the dependent variable?"

Strategy: Know precise definitions. The DV is what is *measured*, not manipulated.

Scenario questions: > "A researcher compares memory performance in quiet vs noisy conditions. What is the IV?"

Strategy: Identify what was manipulated (noise level) vs measured (memory).

"Best" questions: > "Which is the BEST example of random sampling?"

Strategy: Look for the option that most closely matches the textbook definition. "Best" implies comparison.

"EXCEPT" questions: > "All of the following are true about correlational research EXCEPT…"

Strategy: Find the one false statement. Mark these questions carefully — it is easy to get confused.

Interpretation questions: > "A t-test produced p = .03. Which statement is correct?"

Strategy: Know what p-values do and do not mean. Eliminate options that make common misinterpretations.

12.4 Time Management

Pacing

- Note the total number of questions and time available
- Calculate roughly how long per question
- Do not spend too long on any single question
- Mark uncertain questions and return if time permits
- Leave time to review

12.5 When You Are Unsure

1. **Eliminate obviously wrong options** — Even guessing from 2 instead of 4 doubles your odds
2. **Look for absolute language** — Options with "always," "never," "only" are often wrong
3. **Consider what the question is testing** — What concept does this relate to?
4. **Trust your preparation** — Your first instinct, based on study, is often correct
5. **Do not overthink** — Sometimes the straightforward interpretation is correct

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13 Part 6: Quick Reference Tables

13.1 Design Identification

Scenario	Design
Group A gets treatment, Group B does not	Between-subjects
Same participants tested before and after	Within-subjects
Participants matched on IQ, then split into conditions	Matched pairs
Measuring relationship between two variables	Correlational

13.2 Statistical Test Selection

Situation	Test
Comparing two independent groups	Independent t-test
Comparing two related conditions (same participants)	Paired t-test
Examining relationship between two continuous variables	Correlation
Comparing more than two groups	ANOVA (beyond this module)

13.3 Validity Threats

Threat	Definition
Confound	Variable that varies with the IV and could explain results
Selection bias	Non-random assignment leads to non-equivalent groups
Attrition	Participants drop out differentially across conditions
Order effects	Performance affected by sequence of conditions
Demand characteristics	Participants guess the hypothesis and change behaviour
Experimenter effects	Researcher unintentionally influences results

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14 Part 7: What to Expect

14.1 Exam Format

- Multiple-choice questions
- Each question has one correct answer
- Questions cover material from lectures, readings, and labs
- Some questions test recall; others test application

14.2 Content Coverage

The exam will include questions on:

- The scientific method and why it matters
- Variables: IV, DV, confounds, extraneous
- Research designs: between, within, correlational
- Hypotheses: experimental, null, one-tailed, two-tailed
- Sampling methods
- Descriptive statistics: mean, SD
- Inferential statistics: p-values, significance, Type I/II errors
- Reliability and validity
- Research credibility: replication, open science

14.3 On the Day

Practical advice:

1. **Arrive early** — Rushing increases anxiety
2. **Read instructions carefully** — Note how many questions, how much time
3. **Pace yourself** — Do not get stuck on difficult questions

4. **Stay calm** — If you have studied, you are prepared
5. **Trust your preparation** — You know more than you think

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15 Summary

Area	Key Points to Review
Variables	IV (manipulated), DV (measured), confounds
Designs	Between, within, correlational — know the differences
Hypotheses	Experimental vs null, directional vs non-directional
Statistics	Mean, SD, p-values, significance
Credibility	Reliability, validity, replication

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16 Check Your Understanding

Test yourself on these questions:

1. What is the difference between an IV and a DV?
2. When would you use a within-subjects design vs a between-subjects design?
3. What does a p-value actually tell you?
4. What is the difference between reliability and validity?
5. Why does correlation not imply causation?
6. What is pre-registration and why does it matter?
7. What is the difference between Type I and Type II errors?

If you can explain each of these clearly, you are well prepared.

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17 Final Thoughts

Remember

The exam tests understanding, not memorisation. If you understand *why* things work the way they do — why we need control conditions, why p-values are not probabilities of hypotheses being true, why replication matters — you can reason your way through most questions.

Good luck!